

Czech Academy of Sciences–Kanazawa University Workshop  
Shiinoki Cultural Complex, Kanazawa City, Ishikawa, Japan  
November 2–3, 2025

| Time        | Title of Talk / Speaker / Affiliation                                                                                                                                                                             |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 09:50–10:00 | <u>Masato Kimura</u> , Kanazawa University                                                                                                                                                                        |
| 10:00–10:50 | <i>Weak solutions to a full compressible magnetohydrodynamic flow interacting with thermoelastic structure and singular limits</i><br><u>Šárka Nečasová</u> , Institute of Mathematics, Czech Academy of Sciences |
| 10:50–11:05 | <i>Non-axisymmetric tornado-type flow: energy transfer and dynamics</i><br><u>Afifah Maya Iknaningrum</u> , Kanazawa University                                                                                   |
| 11:05–11:20 | Break                                                                                                                                                                                                             |
| 11:20–11:35 | <i>A variable time step Lagrange–Galerkin scheme with second-order accuracy in time for convection-diffusion problems</i><br><u>Yuki Karasawa</u> , Kanazawa University                                           |
| 11:35–11:50 | <i>Dynamics of the wave-pinning model for cell polarity</i><br><u>Taikei Uechi</u> , Kanazawa University                                                                                                          |
| 11:50–12:20 | <i>A rate-independent model of droplet evolution</i><br><u>Norbert Pozar</u> , Kanazawa University                                                                                                                |
| 12:20–14:00 | Lunch Break                                                                                                                                                                                                       |
| 14:00–14:30 | <i>3 topics on many-particle limits</i><br><u>Patrick van Meurs</u> , Kanazawa University                                                                                                                         |
| 14:30–15:20 | <i>Analysis of bi-fluid systems</i><br><u>Martin Kalousek</u> , Institute of Mathematics, Czech Academy of Sciences                                                                                               |
| 15:20–15:35 | Break                                                                                                                                                                                                             |
| 15:35–15:50 | <i>On the memory of the twin vortex computer for an optimized cylinder</i><br><u>Yuma Nakamura</u> , Kanazawa University                                                                                          |
| 15:50–16:40 | <i>On compressible fluids with shear dependent viscosity</i><br><u>Václav Mácha</u> , Institute of Mathematics, Czech Academy of Sciences                                                                         |
| 16:40–16:55 | Break                                                                                                                                                                                                             |
| 16:55–17:45 | <i>On dissipative turbulent solutions to the compressible anisotropic Navier–Stokes equations in unbounded domains</i><br><u>Ondřej Kreml</u> , Institute of Mathematics, Czech Academy of Sciences               |

Table 1: Day 1 Schedule

| Time        | Title of Talk / Speaker / Affiliation                                                                                                                                               |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 09:30–10:20 | <i>Equilibrium state of the 3D MHD equations with an arbitrary geometry</i><br><u>Hideo Kozono</u> , Waseda University & Tohoku University                                          |
| 10:20–10:35 | <i>Parameter identification in elliptic PDEs using the coupled complex boundary method with Tikhonov regularization</i><br><u>Sahat Pandapotan Nainggolan</u> , Kanazawa University |
| 10:35–10:50 | <i>Fracture phase field model with unilateral contact condition: energy dissipation identity and finite element simulations</i><br><u>Oussama Ounissi</u> , Kanazawa University     |
| 10:50–11:05 | Break                                                                                                                                                                               |
| 11:05–11:20 | <i>Lagrangian–Galerkin moving mesh method</i><br><u>Kharisma Surya Putri</u> , Kanazawa University                                                                                  |
| 11:20–12:10 | <i>Numerical modelling of human phonation process</i><br><u>Jan Valášek</u> , Institute of Mathematics, Czech Academy of Sciences                                                   |
| 12:10–14:00 | Lunch Break                                                                                                                                                                         |
| 14:00–14:30 | <i>Well-posedness of the Langmuir film model</i><br><u>Koya Sakakibara</u> , Kanazawa University                                                                                    |
| 14:30–15:20 | <i>Stability of bifurcating patterns in viscous compressible fluids</i><br><u>Yoshiyuki Kagei</u> , Institute of Science Tokyo                                                      |
| 15:20–15:35 | Break                                                                                                                                                                               |
| 15:35–16:50 | <i>TBA</i><br><u>Naohiko Ogawa</u> , Kanazawa University                                                                                                                            |
| 15:50–16:40 | <i>Spatially adaptive stabilized Lagrange–Galerkin schemes for two-fluid flow and fluid–structure interaction problems</i><br><u>Hirofumi Notsu</u> , Kanazawa University           |
| 16:40–16:30 | <u>Hirofumi Notsu</u> , Kanazawa University                                                                                                                                         |
| 16:30–18:00 | Project / Free Discussion                                                                                                                                                           |

Table 2: Day 2 Schedule